Vision-Based Misalignment Diagnosis of the SOFIA Hexagonal Units: Geometric and Learned Estimators

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***Abstract*—Each SOFIA module carries 16 hexagonal plates, each tilted by two servomotors [1]. A plate that does not return to its zero position after a motion is difficult to notice by eye, so this work estimates the two joint angles of every plate from a single overhead photograph and classifies each plate as OK ($< 2^\circ$), REVIEW (2–5°), or MISALIGNED ($> 5^\circ$). Four ArUco markers placed at the height of the plates rectify the photograph, and three estimators are compared on 2000 images rendered in Webots: a training-free geometric method, a small convolutional neural network (CNN), and a pretrained ResNet18. On 300 test images (4800 plates), the CNN reduces the mean absolute error from 0.54° to 0.25° and the maximum error from 74° to 4.4°, while the geometric method remains the most accurate below 5°. A hybrid rule that selects, for each plate, between both estimators achieves the best results, with a mean error of 0.22°, a maximum error of 4.4°, and a state accuracy of 97.2 %. The estimators are integrated in a diagnosis application with a dedicated processing thread, which can also be connected to a live Webots simulation to misalign the plates interactively and compare each diagnosis with the angles measured in the simulation.**

***Index Terms*—Misalignment diagnosis, fiducial markers, homography, convolutional neural networks, Webots, synthetic data, multithreading.**

I. Method

A. Setup and Synthetic Dataset

The module was reproduced in Webots [2] with the real panel layout: 16 plates in four staggered rows, a plate radius of 69 mm, and both joint axes at $L = 58$ mm below the plate. A frame with four ArUco markers [3] (60 mm, dictionary 4×4) surrounds the array, with their centers on a 0.76 m square at the height of the plates. A generator controller set random joint angles, rendered one photograph, and stored the 32 angles as labels. To mimic a hand-placed camera, each image used a camera height of 1.30–1.40 m, tilts of up to $\pm 4^\circ$, a yaw of up to $\pm 8^\circ$, and an offset of up to ± 4 cm. Joint angles were drawn as 5 % at zero, 50 % within $\pm 3^\circ$ (the region of the thresholds), and the rest within $\pm 27^\circ$ (the mechanical limit is $\pm 30^\circ$). Image 0, with every plate at zero, is the reference. The other 1999 images were split *by image* into 1399 for training, 300 for validation, and 300 for testing, so the 16 plates of a photograph never fall in different subsets.

B. Rectification and Crops

The pipeline is shown in Fig. 1. Since the markers lie in the plane of the plates, the homography computed from their four centers maps that plane exactly to a top view at 1 px/mm (860×860 px), independently of the camera pose. In the rectified image, every plate at zero falls on a known pixel, and a crop of 218×218 mm around it is resized to 128×128 px. The same rectified image and crops feed all the estimators.

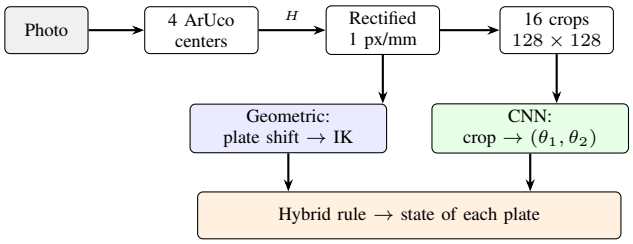


Fig. 1. Diagnosis pipeline. The homography H maps the plane of the plates to a fixed top view, so every plate appears at a known position.

C. Geometric Estimator

The plate orientation is $R_x(\theta_1)R_y(\theta_2)(0,0,L)^\top$, so its center moves in the top view by

$$\Delta v = L \sin\theta_2, \qquad \Delta u = L \cos\theta_2 \sin\theta_1, \tag{1}$$

which is inverted as $\theta_2 = \arcsin(\Delta v/L)$ and $\theta_1 = \arcsin\big(\Delta u/(L\cos\theta_2)\big)$. In a window around each zero position, the image is binarized with Otsu's method [4], a hexagonal template locates the plate by normalized cross-correlation with sub-pixel refinement, and the centroid of the connected bright region replaces that estimate when its area and solidity match a plate. Displacements are measured with respect to the plates in the reference image, which absorbs small assembly errors. The method needs no training; its known limitation is that a tilted plate also drops by up to 1 cm, which perspective converts into a bias at large angles.

D. Learned Estimators

Two regressors map a crop to $(\theta_1, \theta_2)/30^\circ$ in PyTorch [5]: a small CNN trained from scratch (five blocks of two 3×3 convolutions with batch normalization and max pooling, 32–128 channels, 1.28 M parameters) and a ResNet18 [6] pretrained on ImageNet (13.3 M parameters). Since the answer is *where* the plate lies in the crop, both omit the global average pooling and flatten the final 4×4 feature map into the fully connected head. Training used a smooth-L1 loss, AdamW [7] with a one-cycle schedule [8], mixed precision, and batches of 128, for 40 epochs (CNN) or 25 epochs (ResNet18); the epoch with the lowest validation error was kept. Augmentation applied horizontal and vertical flips, changing the sign of θ_1

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TABLE I
TEST RESULTS (300 IMAGES, 4800 PLATES)

	Error (°)			MAE by $\lvert\theta\rvert$ (°)		State	Time
Method	MAE	RMSE	Max	0–5°	$> 15°$	acc.	(ms)
Geometric	0.539	1.927	74.0	**0.13–0.15**	1.301	**97.2 %**	32[a]
Small CNN	0.248	0.347	4.4	0.18–0.19	**0.363**	96.3 %	**12**[b]
ResNet18	0.293	0.409	**4.1**	0.20–0.23	0.433	95.9 %	16[b]
Hybrid	**0.223**	**0.324**	4.4	**0.13–0.15**	**0.363**	**97.2 %**	32[b]

[a]CPU only. [b]Pre-processing on CPU, network on GPU.

or θ_2, respectively, and random gain, gamma, blur, and noise. Translations, rotations, and scaling were excluded because they would change the answer.

E. Hybrid Rule

The two families fail in different ways (Section II), so both are executed and a per-plate rule selects the result: if the geometric and CNN estimates differ by more than $2°$ on either axis, the CNN value is used and the plate is flagged for inspection; otherwise, the geometric value is used when the CNN places the plate within $6°$ of zero, and the CNN value elsewhere.

II. RESULTS

Table I reports the test results on a desktop PC with an AMD Ryzen CPU and an NVIDIA RTX 3060 GPU. The time per photograph includes the marker detection, the rectification, and the estimation of the 16 plates.

Among the individual estimators, the small CNN obtains the lowest overall error, less than half that of the geometric method, and removes its catastrophic failures: the largest error drops from $74°$ to $4.4°$. These failures occur above $23°$, when a strongly tilted plate is confused with the gap next to it. Conversely, the geometric method is the most accurate below $5°$, where the state thresholds lie, which explains its slightly higher state accuracy (97.2 % versus 96.3 %). The pretrained ResNet18, ten times larger, does not outperform the small CNN: features learned for object recognition offer little advantage for measuring a shift of a few pixels, although its validation error was still decreasing in the last epoch. All the estimators are more accurate in θ_2 than in θ_1 (CNN: $0.21°$ versus $0.28°$). The hybrid rule exploits these complementary behaviors: it matches the geometric method below $5°$ and the CNN above $15°$, reaching the lowest mean error ($0.223°$) and the state accuracy of the geometric method, without its catastrophic failures, since its maximum error equals that of the CNN. The geometric value was selected for 2792 of the 4800 plates (58 %). Of the 286 plates flagged for disagreement, 250 (87 %) corresponded to a geometric error above $2°$, so the flag is a useful indicator for inspection. Its cost is time: both estimators run sequentially, for 32 ms per photograph.

III. DIAGNOSIS APPLICATION AND LIVE TESTING

A. Software Organization

The project is divided into three parts. The *application* contains the diagnosis core (geometry of the module, rectification, the three estimators, and the hybrid rule) and the user interface. The *build* part generates and prepares the dataset, trains and evaluates the models, and publishes the selected one. The *results* part stores what the build produces and the application consumes: the published CNN (5 MB), the evaluation reports, and the printable marker sheet. Since the training scripts import the same core as the application, the rectification, the crops, and the input normalization are identical in both, which avoids a mismatch between the evaluated model and the deployed one. When PyTorch or the published model is not available, the application reports it and falls back to the geometric method, which requires only OpenCV.

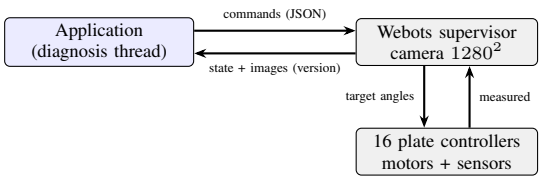


Fig. 2. Live testing loop. The application misaligns the simulated plates and compares each diagnosis with the angles measured in the simulation.

B. Application

The application was implemented in PySide6 following the same threading model as the gesture-recognition interface of SOFIA: a `QThread` reads the image source, loads the model, runs the diagnosis, and draws the result, while the interface thread only displays what it receives through Qt signals, and commands from the interface are queued under a mutex. Three sources are supported: a camera, recorded images, and a live Webots simulation. When the recorded images belong to a Webots session, the stored joint angles are also displayed as ground truth. For each plate, the application draws an arrow whose direction and length follow (θ_1, θ_2) and whose color indicates the state, marks the plates flagged by the hybrid rule, and fills a 4×4 map arranged as in the image. The estimator can be changed at run time, the reference image at zero can be captured from the current source, and each diagnosis can be saved as an image and a CSV file.

C. Live Testing with Webots

To test the complete chain without the physical module, a second world reuses the dataset world (same plates, markers, and camera) with a supervisor that connects to the application through a local TCP socket (Fig. 2). The application sends commands as JSON lines (Table II), and the supervisor forwards the requested angles to the 16 plate controllers, which reach them and report the angles measured by their position sensors. The supervisor transmits two packet types: the state of the plates, with the measured angles, and the 1280×1280 camera images at about five per second.

A diagnosis is only meaningful if the image was taken with the plates at rest in the configuration whose angles are reported. For this reason, every command increments a version number, and an image carries that version only when all the plates have reported being at their target for several

TABLE II
COMMANDS OF THE LIVE TESTING MODE

Command	Effect
Zero	All plates to $(0, 0)$; the next stable image becomes the reference
Random	N random plates to random angles up to a given limit; the rest to zero
Plate	One plate to a given (θ_1, θ_2)
Scene	Camera pose and lights randomized as in the dataset, or restored

consecutive steps; otherwise, it carries zero. The application analyzes automatically, exactly once per version, the first image whose version matches that of a stable state, and displays the measured angles as ground truth together with the error of each plate. In this way, each command produces one diagnosis that can be compared directly with the simulation, and changing the scene tests the robustness to the camera pose without moving the plates. The communication and the automatic analysis were verified against an emulated supervisor; a systematic evaluation in this mode is part of the future work.

IV. CONCLUSIONS

A single overhead photograph with four fiducial markers is sufficient to estimate the 32 joint angles of a SOFIA module: rectification removes the effect of the camera pose, and the resulting top view allows both a training-free geometric estimator and learned regressors. On synthetic data, a small CNN achieves a mean error of 0.25° without large failures, the geometric method is the most accurate near the state thresholds, and a hybrid rule that combines them achieves 0.22° and flags doubtful plates for inspection. The geometric method remains a fallback when no GPU or training data are available. The estimators were integrated in a threaded diagnosis application that also connects to a live Webots simulation, which allows the complete chain, from misalignment to diagnosis, to be tested without the physical module. Future work includes a systematic evaluation in the live mode, the integration of the application as a diagnosis mode of the main SOFIA interface, and validating the estimators on photographs of the physical 4×4 module, including fine-tuning the CNN with real images to close the gap between simulation and reality.

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